

An explanation why “A least squares solution to $\mathbf{Ax} = \mathbf{b}$ is a solution of $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$ ”

Part 1:

For simplicity and without loss of generality, we shall illustrate with a matrix $\mathbf{A}$ with just three columns $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$.

So $\mathbf{A} = (\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3)$ and $\mathbf{A}^T = \begin{pmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \end{pmatrix}$ where $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are regarded as the three rows of $\mathbf{A}^T$.

When we multiply $\mathbf{A}^T$ to a (column) vector $\mathbf{v}$, we have $\mathbf{A}^T\mathbf{v} = \begin{pmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \end{pmatrix} \mathbf{v} = \begin{pmatrix} \mathbf{a}_1 \cdot \mathbf{v} \\ \mathbf{a}_2 \cdot \mathbf{v} \\ \mathbf{a}_3 \cdot \mathbf{v} \end{pmatrix}$. On the right hand side, we take the dot product of each row of $\mathbf{A}$ with the column vector $\mathbf{v}$, which has the same effect as the matrix multiplication.

Part 2:

Now suppose $\mathbf{u}$ is a least squares solution of $\mathbf{Ax} = \mathbf{b}$.

Then $\mathbf{Au}$ is the projection of $\mathbf{b}$ onto the column space of $\mathbf{A}$, which is $\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$.

This means $\mathbf{b} - \mathbf{Au}$ is orthogonal to the column space $\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$. (This is the definition of projection).

So $\mathbf{b} - \mathbf{Au}$ is orthogonal to every vector in the column space, in particular $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$.

In terms of dot product, we have $\mathbf{a}_1 \cdot (\mathbf{b} - \mathbf{Au}) = 0, \mathbf{a}_2 \cdot (\mathbf{b} - \mathbf{Au}) = 0, \mathbf{a}_3 \cdot (\mathbf{b} - \mathbf{Au}) = 0$.

Part 3:

Now to show $\mathbf{u}$ is a solution of $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$:

We need to check $\mathbf{A}^T\mathbf{Au} = \mathbf{A}^T\mathbf{b}$, which is the same as $\mathbf{A}^T\mathbf{b} - \mathbf{A}^T\mathbf{Au} = \mathbf{0}$ or $\mathbf{A}^T(\mathbf{b} - \mathbf{Au}) = \mathbf{0}$. ---(*)

Express left hand side of (*) in terms of the rows of $\mathbf{A}^T$, we have

$$\mathbf{A}^T(\mathbf{b} - \mathbf{Au}) = \begin{pmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \mathbf{a}_3 \end{pmatrix} (\mathbf{b} - \mathbf{Au}) = \begin{pmatrix} \mathbf{a}_1 \cdot (\mathbf{b} - \mathbf{Au}) \\ \mathbf{a}_2 \cdot (\mathbf{b} - \mathbf{Au}) \\ \mathbf{a}_3 \cdot (\mathbf{b} - \mathbf{Au}) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \mathbf{0}, \text{ which is the right hand side of (*).}$$

This shows that $\mathbf{u}$ is a solution of $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$.